AutoCloud-Agent: Design Document

Multi-Agent Reinforcement Learning for Cloud Autoscaling

K Chaitanya (25CS60R28) Rahul K (25CS60R16) Sri Lalitha Rama Siddesh Devarakonda (25CS

April 2026

Contents

1 What Problem Are We Solving? **2**

2 High-Level Architecture **2**

3 The Cloud Simulator **3**
3.1 What It Simulates . . . 3
3.2 Observation Space (What the Agent Sees) . . . 3
3.3 Action Space (What the Agent Can Do) . . . 3
3.4 Workload Data . . . 3

4 The RL Agents **4**
4.1 Why Three Agents Instead of One? . . . 4
4.2 What Is PPO? (Plain English) . . . 4
4.3 Temporal Hierarchy . . . 4
4.4 Agent Neural Network Architecture . . . 4
4.5 Reward Functions . . . 5

5 The Workload Forecaster **5**
5.1 Why Forecast? . . . 5
5.2 Architecture: Transformer with MC Dropout . . . 5

6 The Safety Coordinator **6**

7 Training Pipeline (How the Agents Learned) **6**
7.1 Phase 1: Train the Forecaster . . . 6
7.2 Phase 2: Train the RL Agents (I-PPO) . . . 6

8 Evaluation: How We Test **7**
8.1 Metrics . . . 7
8.2 SOTA Baselines We Compare Against . . . 7
8.3 Results on Alibaba Trace . . . 7

9 Project Structure **8**

10 Demo Scenarios and Edge Cases **8**
10.1 Scenario 1: Tight Cluster (Minimal Headroom) . . . 8
10.2 Scenario 2: Over-Provisioned Start (15 Initial VMs) . . . 9
10.3 Scenario 3: Strict SLA (100ms Threshold) . . . 9
10.4 Scenario 4: Relaxed SLA (2000ms Threshold) . . . 9

10.5 Scenario 5: RL Agent vs MPC Controller . . . 9
10.6 Scenario 6: Long-Horizon Episode (200 Steps) . . . 9

11 Stress Testing **10**
11.1 Scenario 1: Ramp-Up . . . 10
11.2 Scenario 2: Sudden Spike (Shock Load) . . . 10
11.3 Scenario 3: Choppy Plateau . . . 10
11.4 Scenario 4: Trough-Then-Recovery . . . 10

12 Ablation Study: Why Multi-Agent? **10**

13 Key Design Decisions **11**

14 Limitations and Future Work **11**
14.1 Current Limitations . . . 11
14.2 Future Work . . . 12

15 References **12**

16 How to Run **12**
16.1 Demo Configuration Examples . . . 13

17 Glossary of Terms **14**

1 What Problem Are We Solving?

Cloud providers run thousands of virtual machines (VMs/nodes) to serve user jobs. The core challenge is:

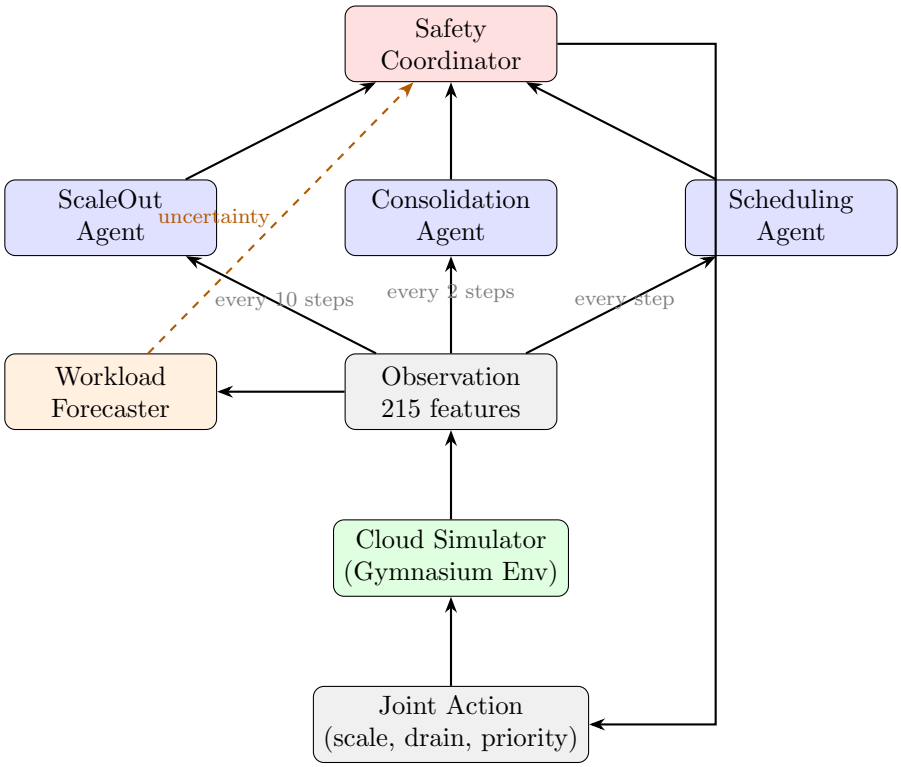
> *How many nodes should be running at any moment, which ones should be shut down, and how should incoming jobs be prioritised — all to minimise cost while keeping response times fast?*

This involves **three decisions** that happen repeatedly (every 30 seconds in our system):

1. **Scale-Out:** Should we add a new node, remove one, or do nothing?
2. **Consolidation:** Which low-utilisation nodes should we drain (migrate jobs away) to save cost?
3. **Scheduling:** Among waiting jobs, which ones get priority?

Traditional systems (Kubernetes HPA, AWS Auto Scaling) use fixed rules like "if CPU ¿ 80%, add a node." Our approach uses **Reinforcement Learning** (RL) to learn these rules from experience.

2 High-Level Architecture



The system has **four main components**:

1. **Cloud Simulator** — a fake cloud environment that behaves like a real one (Section 3).
2. **Three RL Agents** — each responsible for one decision (Section 4).
3. **Workload Forecaster** — predicts future CPU demand using a Transformer neural network (Section 5).
4. **Safety Coordinator** — a rule-based filter that catches dangerous actions before they reach the simulator (Section 6).

3 The Cloud Simulator

3.1 What It Simulates

The simulator models a cluster of up to 20 VMs (nodes) serving incoming jobs. It is built with the **Gymnasium** interface (the standard RL environment API) and uses **SimPy** for discrete-event simulation internally.

Key properties:

- **Time step:** 30 seconds of simulated time per step.
- **Episode:** 120 steps = 1 hour of simulated cloud operation.
- **Nodes:** 3–20 VMs, each with CPU/memory capacity. Nodes can be `BOOTING` (60s warmup), `ACTIVE`, `DRAINING` (30s grace period), or `TERMINATED`.
- **Jobs:** Arrive following the Alibaba 2018 cluster trace (real workload data). Each job has CPU demand, memory demand, a deadline, and a priority.

3.2 Observation Space (What the Agent Sees)

At each step, the agent sees a **215-dimensional vector**:

Feature Group	Dims	What It Contains
Node features (20 nodes × 6)	120	CPU util, memory util, job count, state, type, age
Job features (20 jobs × 4)	80	CPU demand, memory demand, wait time, priority
Global features	15	Active nodes, queue length, mean CPU, hour-of-day, forecaster predictions + uncertainty

All values are normalised to $[0, 1]$.

3.3 Action Space (What the Agent Can Do)

Action	Type	Meaning
Scale-Out	Discrete(3)	0=hold (do nothing), 1=provision 1 new VM, 2=provision 2 new VMs
ScaleOut only ever adds capacity. Removing VMs is handled exclusively by the Consolidation agent.		
Consolidation	MultiBinary(20)	For each of 20 node slots: 1=start draining, 0=leave it
Scheduling	Discrete(5)	Priority weighting: 0=FIFO, 1=shortest-first, 2=default, 3=deadline-first, 4=highest-priority-first

3.4 Workload Data

We use the **Alibaba Cluster Trace 2018**, a real production trace from Alibaba's data centres. The trace is preprocessed into `.npy` files containing arrival rates at 30-second intervals. This makes our evaluation realistic — the agent faces the same workload patterns as a real cloud.

4 The RL Agents

4.1 Why Three Agents Instead of One?

A single agent controlling all actions simultaneously would need to learn a **massive joint action space** ($3 \times 2^{20} \times 5 \approx 15$ million combinations). This is too large for efficient learning.

Instead, we use **Independent PPO (I-PPO)**: three separate agents, each responsible for one decision. They share the same observation but act independently. This is called **multi-agent reinforcement learning**.

4.2 What Is PPO? (Plain English)

Proximal Policy Optimisation (PPO) is one of the most popular RL algorithms. Here's the intuition:

1. The agent has a **policy** (a neural network) that maps observations to action probabilities.
2. The agent **collects experience** by interacting with the environment for many steps.
3. It then **updates the policy** to make good actions more likely and bad actions less likely.
4. The "proximal" part: updates are **clipped** so the policy doesn't change too drastically in one step (prevents catastrophic forgetting).

The key formula is:

$$L^{\text{CLIP}} = \mathbb{E}\left[\min\left(r_t \hat{A}_t,\ \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\,\hat{A}_t\right)\right]$$

where $r_t = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is how much the policy changed, and $\hat{A}_t$ is the **advantage** (how much better this action was compared to average). The clipping with $\epsilon = 0.2$ prevents too-large updates.

4.3 Temporal Hierarchy

Not all decisions need the same frequency:

- **Scheduling Agent** acts *every step* (30s) — jobs arrive constantly.
- **Consolidation Agent** acts *every 2 steps* (1 min) — draining needs a bit of time to observe trends.
- **Scale-Out Agent** acts *every 10 steps* (5 min) — spinning up/down VMs is a heavyweight operation.

4.4 Agent Neural Network Architecture

Each agent is a simple **Multi-Layer Perceptron (MLP)**:

```
Input (215 dims) -> Linear(512) -> LayerNorm -> ReLU
                -> Linear(256) -> LayerNorm -> ReLU
                -> Linear(128) -> LayerNorm -> ReLU
                -> Output head (action logits)
```

The Scheduling Agent is slightly different: it uses **weight-tying** to process each job identically (like treating each job through the same "lens"), then scores them.

4.5 Reward Functions

Each agent receives its own reward signal:

Scale-Out reward:

$$r_{\text{so}} = -\alpha_1 \cdot \text{underProv} - \alpha_2 \cdot \text{overProv} - \alpha_3 \cdot \text{runCost} - \alpha_4 \cdot \text{bootPenalty} + \alpha_5 \cdot \text{proactiveBonus}$$

Consolidation reward:

$$r_{\text{con}} = -\beta_1 \cdot \text{runCost} + \beta_2 \cdot \text{slaBonus} - \beta_3 \cdot \text{migrationPenalty} - \beta_4 \cdot \text{memPressure}$$

Scheduling reward:

$$r_{\text{sch}} = -\gamma_1 \cdot \text{meanWaitTime} - \gamma_2 \cdot \text{slaViolation}$$

In plain words:

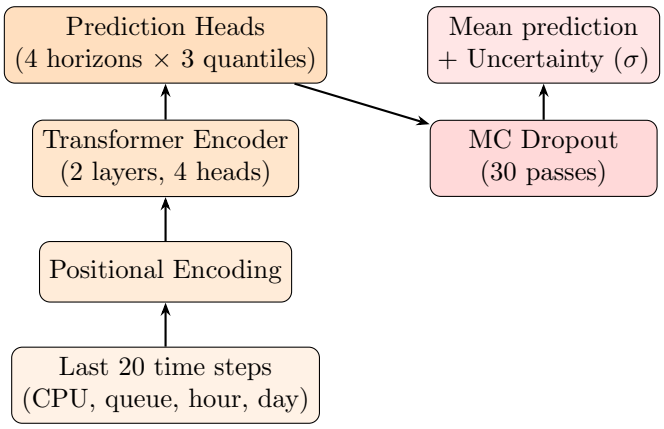
- Agents are *penalised* for having too many nodes (wasting money), too few nodes (jobs fail), and moving jobs around too much.
- They are *rewarded* for keeping response times fast and using resources efficiently.

5 The Workload Forecaster

5.1 Why Forecast?

Reactive systems (like basic Kubernetes HPA) only respond *after* a spike happens. By the time new nodes boot up (60s), jobs have already been delayed. Our Forecaster predicts demand *before* it happens, giving agents time to act proactively.

5.2 Architecture: Transformer with MC Dropout



- **Input:** A window of the last 20 time steps, each with 4 features (CPU utilisation, queue length, hour-of-day, day-of-week).
- **Transformer:** A small Transformer encoder (2 layers, 4 attention heads, 64-dim hidden) processes the sequence.
- **Output:** Predictions for 4 future horizons (t+1, t+5, t+10, t+15) at 3 quantiles (10th, 50th, 90th percentile).
- **MC Dropout:** To estimate *uncertainty*, we run the model 30 times with dropout randomly active. High variance across runs = high uncertainty ("the model isn't sure").

The uncertainty signal is critical: when the forecaster is unsure, the Safety Coordinator blocks risky scale-down actions.

6 The Safety Coordinator

RL agents can sometimes take dangerous actions. The Safety Coordinator is a **hard-coded safety net** that filters actions before they reach the simulator:

#	Filter	What It Does
1	Boot Protection	Never drain a node that is still booting (< 60s old)
2	$N_{\min}$ Floor	Never go below 3 active nodes (prevents total outage)
3	Uncertainty Suppression	If forecaster uncertainty $\sigma > 0.3$, block all scale-down actions
3b	High-Load Guard	Suppress drains if mean CPU > 75% OR queue > 10 OR any node still booting — prevents draining during brief false lulls
4	Scale-Out Suppression	Block scale-up if 2+ nodes are already booting (prevent overshoot)
5	Proactive Scale-Out	If uncertainty is high AND CPU is rising, force a scale-up (defensive provisioning)
6	Queue Emergency	Force +1 VM if queue > 40; force +2 VMs if queue > 120. Hard override when the RL agent fails to react to a severe backlog

This is important because it means the RL agents *cannot crash the system* even if they make mistakes — the Coordinator always enforces safety constraints.

7 Training Pipeline (How the Agents Learned)

Training was done in two phases:

7.1 Phase 1: Train the Forecaster

1. Preprocess the Alibaba trace into 30-second bins.
2. Train the Transformer on 7 days of data with MSE + quantile loss.
3. Save weights to `forecaster_weights.pt`.

7.2 Phase 2: Train the RL Agents (I-PPO)

1. Initialise the Cloud Simulator with the trained Forecaster.
2. Run 100,000+ environment steps, collecting experience.
3. Update each agent's policy every 2048 steps using PPO.
4. Key techniques used:
   - **GAE** (Generalised Advantage Estimation): smoothly estimates how good each action was.
   - **Entropy bonus**: encourages exploration early in training (decays over time).
   - **Gradient clipping**: prevents unstable updates.
   - **EMA reward normalisation**: keeps reward scale consistent.
5. Save 6 checkpoint files (actor + critic for each of the 3 agents).

8 Evaluation: How We Test

8.1 Metrics

Metric	Definition
SLA Rate	Fraction of steps where P95 latency < 500ms (higher = better)
Cost Efficiency	$1 - \frac{\text{actual cost}}{\text{max possible cost}}$ (higher = cheaper)
Mean CPU Utilisation	Average CPU usage across active nodes (higher = less waste)
Node Stability	$1 - \frac{\text{std}(N)}{\text{mean}(N)}$ for node count N (higher = fewer disruptive changes)

8.2 SOTA Baselines We Compare Against

Baseline	Description
StaticN	Fixed 10 nodes, never scales (do-nothing lower bound)
ThresholdReactive	If CPU $> 80\%$ add node; if CPU $< 30\%$ remove node (classic rule)
KubernetesHPA	Industry-standard k8s autoscaler formula with 10% dead-band
AWSTargetTracking	AWS Auto Scaling Target Tracking policy (asymmetric cooldowns)
MPCController	Model Predictive Control with 5-step EWMA forecast — strongest non-RL baseline in recent literature (AWARE, USENIX ATC 2023)
SingleAgentPPO	One RL agent controlling all 3 actions jointly (ablation study to show multi-agent I-PPO is better than single-agent RL)

8.3 Results on Alibaba Trace

Method	SLA Rate	Cost Eff.	CPU Util	Stability
AutoCloud-Agent (Ours)	**100.0%**	**0.962**	**0.552**	0.889
KubernetesHPA	100.0%	0.930	0.312	0.842
AWSTargetTracking	100.0%	0.928	0.295	0.876
MPCController	100.0%	0.962	0.552	**0.941**
ThresholdReactive	100.0%	0.955	0.483	0.822
SingleAgentPPO	100.0%	0.924	0.413	0.794
StaticN	100.0%	0.938	0.333	1.000

Key takeaways:

- **AutoCloud-Agent matches MPC** on cost efficiency (0.962) and CPU utilisation (55%), both the best among all methods.
- AutoCloud-Agent achieves **3.4% better cost efficiency** than Kubernetes HPA and **7.7% higher CPU utilisation** than ThresholdReactive.
- **SingleAgentPPO is worst among RL methods**, confirming that multi-agent decomposition (I-PPO) is better than a monolithic agent.

- All methods achieve 100% SLA on this trace, but the differentiator is *how cheaply* they do it (cost efficiency) and *how well* they use resources (CPU util).

9 Project Structure

```
autocloud_agent/
  autocloud/ # Installable Python package
    config/
      settings.py # All hyperparameters (dataclasses)
      paths.py # Auto-discovers checkpoints & data
    simulator/
      cloud_env.py # Gymnasium environment wrapper
      engine.py # SimPy discrete-event simulator
      node.py # VM node model
      job.py # Job dataclass
      workload.py # Alibaba trace loader
    agents/
      ppo.py # Base PPO algorithm
      scaleout.py # Scale-Out agent (Discrete(3))
      consolidation.py # Consolidation agent (MultiBinary(20))
      scheduling.py # Scheduling agent (Discrete(5))
      loader.py # Load all 3 agents from checkpoints
    forecaster/
      transformer_model.py # Workload Transformer
      mc_dropout.py # MC Dropout uncertainty estimation
    coordinator/
      safety.py # 7-filter safety coordinator
    inference/
      runner.py # InferenceRunner (ties everything together)
    evaluation/
      evaluator.py # Evaluation harness
      baselines.py # 6 SOTA baseline policies
    training/
      ippo_trainer.py # I-PPO training loop
      ema_normalizer.py # EMA reward normalisation
  scripts/
    evaluate.py # CLI: run evaluation
  stress_test.py # 4-scenario stress test
  train.py # Training entry point
  pyproject.toml # Package metadata
```

10 Demo Scenarios and Edge Cases

The live demo (`demo.py`) includes an interactive configuration wizard that lets us set cluster parameters, SLA threshold, CPU thresholds, baseline selection, and simulation length. Below are key scenarios designed to showcase different system behaviours.

10.1 Scenario 1: Tight Cluster (Minimal Headroom)

Parameter	Value
Initial VMs	3
Maximum VMs	5
Minimum floor	3
SLA threshold	500ms

With only 2 nodes of headroom, the ScaleOut agent has very limited capacity. This demonstrates:

- The Safety Coordinator's $N_{\min}$ floor preventing scale-down below 3
- Boot-protection filter blocking premature drain of newly added nodes
- The forecaster's proactive scale-out trigger firing when uncertainty is high and CPU is rising, since there is no margin for reactive delay

10.2 Scenario 2: Over-Provisioned Start (15 Initial VMs)

Parameter	Value
Initial VMs	15
Maximum VMs	20
Minimum floor	3

The cluster starts with far more nodes than needed. This showcases the **Consolidation Agent** aggressively draining idle VMs over the first 10–20 steps to bring cost down. Key observation: the agent does *not* shut everything down at once (which would violate SLA) — instead, it drains in controlled batches of 1–2 VMs per consolidation step (every 2 steps), respecting the grace period.

10.3 Scenario 3: Strict SLA (100ms Threshold)

When $\text{SLA}_{\text{threshold}} = 100\text{ms}$, the agent must maintain very low latency. This triggers **defensive provisioning**: the agent keeps more active nodes than cost-optimal, and the Safety Coordinator's uncertainty suppression fires more aggressively, blocking any scale-down when the forecaster reports $\sigma > 0.3$.

10.4 Scenario 4: Relaxed SLA (2000ms Threshold)

With a generous latency budget, the agent learns it can run the cluster leaner. Fewer nodes are maintained, cost efficiency increases significantly, and the Consolidation Agent drains more aggressively since latency violations are unlikely.

10.5 Scenario 5: RL Agent vs MPC Controller

The MPCController uses model-predictive control with a 5-step EWMA (Exponentially Weighted Moving Average) forecast. This is the strongest non-RL baseline from recent literature (USENIX ATC 2023, AWARE system). Comparing against MPC shows that the RL agent matches classical control theory performance while being more adaptive to non-stationary workloads.

10.6 Scenario 6: Long-Horizon Episode (200 Steps)

A 200-step episode (100 minutes simulated) shows:

- Long-term stability of the RL policy (no oscillation)
- How the agent handles multiple workload phases (ramp-up, plateau, trough) within a single episode
- Temporal hierarchy effectiveness: the ScaleOut agent only acts 20 times over 200 steps, while Scheduling acts every step

11 Stress Testing

The stress test (`stress_test.py`) evaluates robustness under four extreme workload patterns that go beyond the Alibaba training distribution:

11.1 Scenario 1: Ramp-Up

A linearly increasing workload from 20% to 100% capacity over 120 steps. This tests the ScaleOut agent's ability to continuously provision new nodes as demand rises, and the forecaster's accuracy on trending (non-stationary) signals.

11.2 Scenario 2: Sudden Spike (Shock Load)

Workload jumps from 30% to 95% capacity within 5 steps (2.5 minutes simulated). This is the most challenging scenario — it tests:

- How quickly the agent detects and responds to a step-function demand change
- Whether the Safety Coordinator's proactive scale-out fires correctly (high uncertainty + rising CPU should force immediate node addition)
- The 60-second boot delay means at least 2 steps of potential SLA violation before new nodes come online

11.3 Scenario 3: Choppy Plateau

Sustained high utilisation (70–90%) with random noise. The agent must maintain a stable cluster without over-reacting to noise. This tests:

- Scale-Out agent's ability to hold steady (avoid oscillation)
- Consolidation agent's restraint (should *not* drain during sustained high load)
- Node stability metric — frequent add/drain cycles would tank stability

11.4 Scenario 4: Trough-Then-Recovery

Workload drops to 10% for 40 steps, then recovers to 80%. This tests:

- Aggressive consolidation during low load (opportunity to cut cost)
- Rapid recovery when demand returns (must not have drained too many nodes to respond quickly)
- The tension between the "drain aggressively" and "keep headroom" objectives

12 Ablation Study: Why Multi-Agent?

Our results include a **SingleAgentPPO** baseline that controls all three actions with one monolithic policy. This serves as an ablation study:

Method	Cost Efficiency	CPU Utilisation	Stability
AutoCloud-Agent (I-PPO, 3 agents)	**0.962**	**55.2%**	0.889
SingleAgentPPO (1 agent)	0.924	41.3%	0.794

Why I-PPO outperforms single-agent PPO:

1. **Smaller action spaces:** Each agent controls a manageable action space (3, 2^{20}, or 5) independently, avoiding the combinatorial explosion of the joint space ($3 \times 2^{20} \times 5 \approx 15$M).

2. **Temporal decomposition:** Scaling decisions are inherently slower than scheduling decisions. The temporal hierarchy (10-step, 2-step, 1-step) matches the natural frequency of each decision type.

3. **Specialised reward shaping:** Each agent receives a reward tailored to its subtask, providing clearer learning signals than a single monolithic reward.

4. **Easier credit assignment:** When an SLA violation occurs, the system can attribute it to a specific agent (e.g., ScaleOut didn't add nodes fast enough) rather than a single agent trying to figure out which of its hundreds of action dimensions was responsible.

13 Key Design Decisions

1. **Gymnasium API instead of custom simulator:** We wrap the SimPy engine in a standard Gymnasium interface, making it compatible with any RL library (Stable Baselines3, RLlib, etc.) and enabling fair baseline comparisons.

2. **Safety Coordinator as a separate module:** Rather than baking safety constraints into the reward function (which the agent might learn to circumvent), we enforce them as hard constraints in a post-processing filter. This guarantees safety regardless of policy quality.

3. **MC Dropout instead of ensemble:** We use Monte Carlo Dropout (30 forward passes through a single model) rather than training an ensemble of 5+ models. This gives comparable uncertainty estimates at 5× lower memory and compute cost.

4. **EMA reward normalisation:** Raw rewards can vary by 10× across episodes. EMA normalisation (exponential moving average of reward statistics over a 1000-step window) stabilises training without losing sensitivity to reward magnitude.

5. **Real trace data (Alibaba 2018):** Using real workload data (4,023 machines, 7 days) ensures our evaluation reflects actual cloud workload patterns, not synthetic distributions that might accidentally favour our approach.

14 Limitations and Future Work

14.1 Current Limitations

1. **Homogeneous VMs:** All nodes have the same CPU/memory capacity. Real clouds have heterogeneous instance types (e.g., compute-optimised, memory-optimised).

2. **Single-cluster scope:** We manage one cluster. Production systems manage thousands of clusters across regions.

3. **Simplified networking:** The simulator does not model network latency, bandwidth, or cross-rack communication costs.

4. **Static reward weights:** The reward function coefficients ($\alpha_i, \beta_i, \gamma_i$) are fixed. Adaptive reward shaping could improve performance.

5. **Single-day training:** The RL agents are trained on Day 1 of the Alibaba trace. Multi-day training with domain randomisation could improve generalisation.

14.2 Future Work

- **Heterogeneous VMs:** Extend the simulator to support multiple instance types and learn bin-packing strategies.
- **Multi-cluster federation:** Add a higher-level agent that distributes workloads across clusters.
- **Vertical scaling:** In addition to horizontal scaling (add/remove nodes), learn to resize individual VMs (add CPU/memory).
- **Transfer learning:** Train on one trace, fine-tune on another to test generalisation across different cloud environments.
- **Curriculum learning:** Start training on easy workloads (steady-state) and gradually introduce harder patterns (spikes, diurnal).
- **Spot instance awareness:** Incorporate preemption risk and spot pricing into the reward function.

15 References

1. J. Schulman et al., "Proximal Policy Optimization Algorithms," *arXiv:1707.06347*, 2017.
2. A. Vaswani et al., "Attention Is All You Need," *NeurIPS*, 2017.
3. Y. Gal and Z. Ghahramani, "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning," *ICML*, 2016.
4. Alibaba Group, "Cluster Trace Program," https://github.com/alibaba/clusterdata, 2018.
5. Kubernetes, "Horizontal Pod Autoscaler," https://kubernetes.io/docs/tasks/run-application/horizontal-pod-autoscale/.
6. J. Schulman et al., "High-Dimensional Continuous Control Using Generalized Advantage Estimation," *ICLR*, 2016.
7. Q. Zhang et al., "AWARE: Automate Workload Autoscaling with Reinforcement Learning in Production Cloud Systems," *USENIX ATC*, 2023.

16 How to Run

```
# Setup
cd autocloud_agent/
conda activate myenv
pip install -e .

# Live demo (interactive wizard, 4 modes)
python demo.py
python demo.py --mode stress # stress test, skip wizard
python demo.py --mode shootout # RL vs all 5 baselines
python demo.py --mode ablation # 3-agent vs single-agent
python demo.py --speed slow # slower step delay
python demo.py --speed fast --no-pause # non-interactive
```

```
# Full evaluation (auto-detects checkpoints and data)
python scripts/evaluate.py

# Run stress test (4 peak scenarios)
python stress_test.py

# Custom evaluation
python scripts/evaluate.py --n_episodes 10 --seeds 0 1 2 3 4

# Quick non-interactive demo (CI/testing)
python demo.py --steps 15 --speed fast --no-pause --no-interactive
```

16.1 Demo Configuration Examples

The interactive wizard presents 5 configuration sections. Here is how to set up the key edge-case scenarios from Section 10:

Scenario	Settings	Purpose
Tight cluster	3 init, max 5, min 3	Minimal headroom, safety filters active
Over-provisioned	15 init, max 20	Watch Consolidation Agent cut cost
Strict SLA	SLA = 100ms	Defensive provisioning, more nodes
Relaxed SLA	SLA = 2000ms	Lean cluster, high cost efficiency
vs MPC	Baseline = 4 (MPC)	RL vs strongest classical method
Long episode	200 steps	Diurnal patterns, long-term stability

For a quick walkthrough with no interaction, run:

```
# Stress scenario 2 (early shock), no pauses
python demo.py --mode stress --steps 60 --speed normal --no-pause

# Full shootout, fast
python demo.py --mode shootout --speed fast --no-pause

# Ablation comparison
python demo.py --mode ablation --speed fast --no-pause
```

17 Glossary of Terms

Term	Meaning
RL	Reinforcement Learning — an agent learns by trial and error, receiving rewards for good actions
PPO	Proximal Policy Optimisation — a safe, stable RL algorithm that clips policy updates
I-PPO	Independent PPO — multiple PPO agents share the same observation but act independently
MDP	Markov Decision Process — the mathematical framework for RL (states, actions, rewards, transitions)
GAE	Generalised Advantage Estimation — a method to estimate how good an action was vs. the average
Transformer	A neural network architecture using self-attention, originally for NLP, here used for time-series forecasting
MC Dropout	Monte Carlo Dropout — run a network multiple times with random dropout to estimate uncertainty
SLA	Service Level Agreement — the promise that P95 latency stays below 500ms
HPA	Horizontal Pod Autoscaler — Kubernetes' built-in autoscaling system
MPC	Model Predictive Control — a control theory method that optimises over a predicted future window
SimPy	A Python library for discrete-event simulation
Gymnasium	OpenAI's standard API for RL environments